# CS532 Quiz 3: Stereo

**Name:** _____

- **Q1:** For a rectified stereo setup with baseline equal to 10 cm. and shared focal of 50 pixels. If the true disparity for a given correspondence is 25 pixels. What is the depth error if a miscalculate the disparity by 1 pixel?

  **Solution:**

  We have the equation relating disparity and depth as $Z = \frac{FB}{D}$, with $Z$ being the depth.

  This will give us a true depth of 20cm for a true disparity of 25px.

  To calculate the depth error, since this is a small disparity error, we can linearize around this disparity (use taylor series approx.) to get:

$$\mathbf{\Delta Z = \frac{FB}{D^2}\Delta D = \frac{50 \times 10}{25^2}1 = 0.8cm}$$

- **Q2:** Consider the same stereo setup as before. How would you restrict the depth range for disparity search to be within 1 to 5 meters?

  **Solution:**

  We can get the minimum disparity as a function of max depth (5m), and max disparity as a function of min. depth (1m):

$$\mathbf{D}_{\text{min}} = \frac{\mathbf{50 \times 0.1}}{\mathbf{5}} = \mathbf{1}\text{px}$$

$$\mathbf{D}_{\text{max}} = \frac{\mathbf{50 \times 0.1}}{\mathbf{1}} = \mathbf{5}\text{px}$$